Exp 4: Qualitative Case Vignettes — Dissecting High-Disagreement Edge Cases (Fine-Grained Flora Grounding, Hydrological Ambiguity, and Prompt-Induced Reasoning Drift)

Case A: Tier 3 Nature — Hup Pa Tat Sinkhole Cave (Uthai Thani, Thailand)

Image: images08.jpg | Ground Truth: Hup Pa Tat | Domain: Karst Sinkhole & Endemic Flora | Gemini: 5/5 (100%) vs. All Others: 0/5 (0%)

Key Visual Nuance: Enclosed limestone karst sinkhole valley; dense canopy of ancient Arenga westerhoutii (Tad) palms; wooden boardwalk walkway.

✓ Gemini 3.8 Flash

Pred: Hup Pa Tat Cave

"Enclosed karst sinkhole valley; dense canopy of ancient Arenga westerhoutii (Tad) palms; sheer cliff walls."

✗ Claude Sonnet 5

Pred: Batu Caves (Malaysia)

"Limestone karst cliff formation; tropical rainforest vegetation with palm trees; cave-like rock structure."

✗ GPT-5.4

Pred: Undetermined tropical cave

"Dense tropical rainforest with palms; no distinctive signage or unique architecture; abstains from specific name."

✗ GPT-5.4 Mini

Pred: Maya Cave (Koh Phi Phi)

"Tropical cave/forest; massive limestone cliff walls with dense jungle, characteristic of Railay area."

✗ Qwen 2.5 VL 72B

Pred: Gunung Mulu Nat. Park (MY)

"Large limestone cliffs surrounded by dense tropical vegetation match features typical of Gunung Mulu."

✗ Llama 4 Maverick

Pred: Phraya Nakhon Cave (TH)

"Large rock formation with hole allowing sunlight to enter, surrounded by lush greenery and sandy path."

Scientific Insight: Fine-grained endemic botanical grounding. Only Gemini identifies Arenga westerhoutii (Tad palms) endemic to Uthai Thani; all competitors suffer from geographic substitution into overseas or generic tropical caves.

Case B: Tier 3 Nature — Haew Narok Waterfall (Khao Yai National Park, Thailand)

Image: images13.jpg | Ground Truth: Haew Narok Waterfall | Domain: Hydrological Landscape | Gemini: 5/5 (100%) vs. All Others: 0/5 (0%)

Key Visual Nuance: Three-tier vertical ravine plunge into dark chasm; sheer reddish-brown rock face; complete lack of human-made infrastructure.

✓ Gemini 3.8 Flash

Pred: Haew Narok Waterfall

"Tall vertical waterfall plunging into deep ravine pool; sheer reddish-brown rock cliff faces; dense canopy."

✗ Claude Sonnet 5

Pred: Waterfall (unidentified)

"Tall narrow waterfall cascading over rocky cliff; dense green vegetation; dark plunge pool; lacks markers."

✗ GPT-5.4

Pred: Unknown waterfall

"Single-drop waterfall plunging into dark plunge pool; sheer rocky cliff face; no unique signage or architecture."

✗ GPT-5.4 Mini

Pred: Unknown waterfall

"Tall narrow waterfall dropping over rocky cliff; no distinctive signage, buildings, or uniquely identifying landmarks."

✗ Qwen 2.5 VL 72B

Pred: Kanger Ghati Nat. Park (IN)

"Waterfall distinctive shape and surrounding dense greenery match features typical of Kanger Ghati Park."

✗ Llama 4 Maverick

Pred: Mutawintji Falls (AU)

"Rocky cliff face; waterfall cascading into pool surrounded by lush greenery; distinct landscape and water flow."

Scientific Insight: Natural hydrological ambiguity boundary. Commercial frontier models (Claude, GPT-5.4) appropriately abstain due to lack of man-made anchors, while open-weights models hallucinate distant continents (India, Australia).

Case C: Tier 4 Casual UGC — 57th Street – 7th Avenue Subway Station (MTA, New York City)

Image: 11.jpg | Ground Truth: 57th St – 7th Ave | Domain: Casual Crowdsourced Photo | Prompt Shift: Reasoning Drift & Naive OCR Trap

Key Visual Nuance: Angled smartphone photo of subway mezzanine; overhead exit sign reads '55 Street & 7 Ave', while platform column tiles display '57'.

✓ Gemini 3.8 Flash (P1–P5)

Pred: 57th Street–7th Ave

"Contextual fusion: exit sign reads '55 St' but platform column signage clearly verifies station stop '57'."

✓ Claude Sonnet 5 (P1–P5)

Pred: 57th Street & 7th Ave

"Directional sign indicates 55 St exit, but column marker '57' correctly disambiguates station identity."

✓ GPT-5.4 (P1–P5)

Pred: 57 Street–7 Ave

"Overhead sign reads '55 St' but multiple platform columns marked '57' conclusively match NYC Subway 57 St."

✗ GPT-5.4 Mini (CoT Drift)

Pred: 57 St (P1) → 145th St (P3)

"Prompt Fragility: P1/P2 correct, but Chain-of-Thought (P3) overthinks and drifts into 145th Street–Lenox Ave!"

✗ Qwen 2.5 VL (Naive OCR)

Pred: 55th Street & 7th Ave

"Superficial OCR: Blindly transcribed overhead exit sign '55 St' and fabricated a non-existent station name."

✗ Llama 4 (Naive OCR)

Pred: 55 Street & 7 Ave

"Superficial OCR: Read '55 St' header without verifying against New York City transit station geography."

Scientific Insight: Contextual multi-cue disambiguation vs. superficial OCR shortcuts. Exposes "Overthinking Drift" in lightweight models under Chain-of-Thought (P3 CoT) and naive visual reading in open-weights models.